

# **Student Performance**

Data Cleaning & Transformation

Week 7

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BSC DATA SCIENCE & ARTIFICIAL INTELLIGENCE

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# 1. Introduction

## 1.1 Aim

The aim of this exercise is to import the Student Performance in Exams dataset into Power BI Desktop, clean and transform it using the Power Query Editor, derive new analytical attributes (Total Marks, Average Mark, Grade and a Pass/Fail Result), reshape the subject scores with the Unpivot transformation, prepare a small relational data model, and create basic DAX measures so that the dataset is ready for visualisation.

## 1.2 Objectives

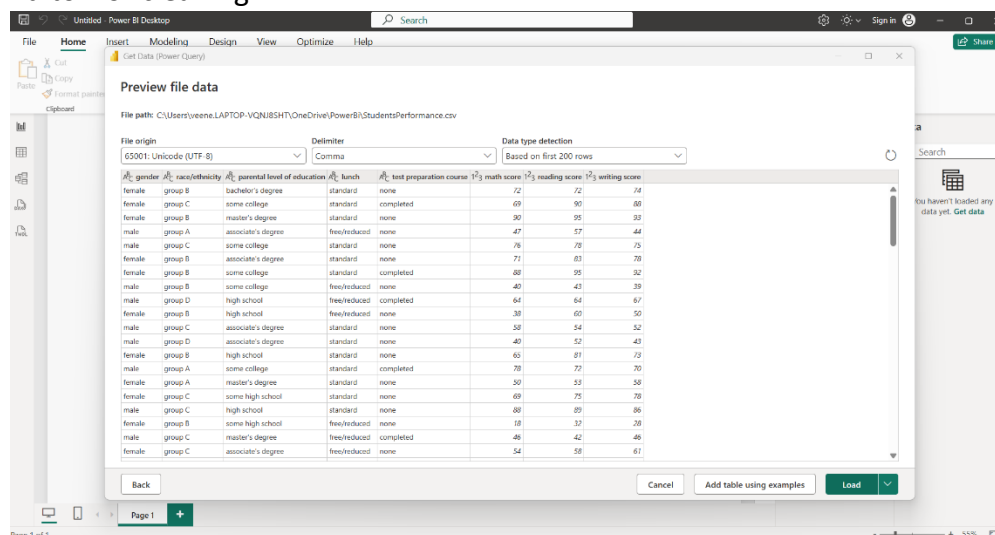
- Import the dataset and open it in the Power Query Editor.
- Explore the columns, rows and data types, and check for missing values, errors and duplicates.
- Clean the dataset: correct data types, standardise categorical values and rename columns clearly.
- Create calculated columns: **Total Marks**, **Average Mark** and a conditional **Grade**.
- Create a **Pass/Fail** classification using a conditional column.
- Reshape the Math, Reading and Writing scores into a long format using **Unpivot**.
- Load the data and create DAX measures: Average Math, Reading and Writing Score, Overall Average Score and Overall Pass %.
- Validate the results against the original dataset.

# 2. Importing and Exploring the Data

## 2.1 Importing the Dataset

The dataset was brought into Power BI Desktop with the following procedure:

1. Open Power BI Desktop and start a new blank report.
2. On the Home ribbon choose Get data → Text/CSV (or Excel workbook if the data is stored as .xlsx) and browse to the Student Performance .
3. Click Transform Data (instead of Load) so that the data opens in the Power Query Editor for cleaning.



This shows the dataset immediately after import. The Queries pane on the left lists a single query named Students Performance.

## 2.2 Exploring and Inspecting the Data

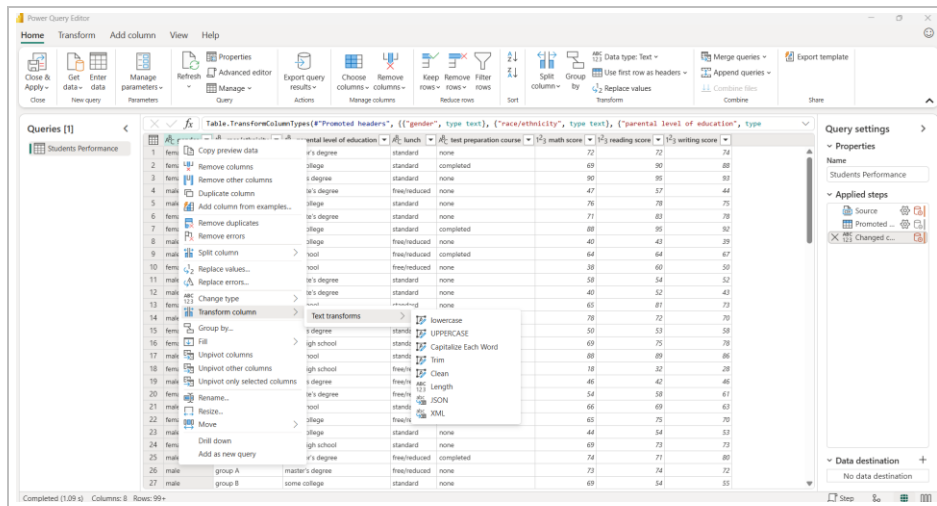
Before changing anything, the structure and quality of the data were reviewed. The data type icon at the left of each column header makes this quick: ABC marks a text column and 123 marks a whole-number column. As the Figure shows, gender, race/ethnicity, parental level of education, lunch and test preparation course are text columns, while math score, reading score and writing score are whole numbers. This Table lists the checks that were carried out.

|    | gender | race/ethnicity | parental level of education | lunch        | test preparation course | math score | reading score | writing score |
|----|--------|----------------|-----------------------------|--------------|-------------------------|------------|---------------|---------------|
| 1  | female | group B        | bachelor's degree           | standard     | none                    | 72         |               |               |
| 2  | female | group C        | some college                | standard     | completed               | 69         |               |               |
| 3  | female | group B        | master's degree             | standard     | none                    | 90         |               |               |
| 4  | male   | group A        | associate's degree          | free/reduced | none                    | 47         |               |               |
| 5  | male   | group C        | some college                | standard     | none                    | 76         |               |               |
| 6  | female | group B        | associate's degree          | standard     | none                    | 71         |               |               |
| 7  | female | group B        | some college                | standard     | completed               | 88         |               |               |
| 8  | male   | group B        | some college                | free/reduced | none                    | 40         |               |               |
| 9  | male   | group D        | high school                 | free/reduced | completed               | 64         |               |               |
| 10 | female | group B        | high school                 | free/reduced | none                    | 38         |               |               |
| 11 | male   | group C        | associate's degree          | standard     | none                    | 58         |               |               |
| 12 | male   | group D        | associate's degree          | standard     | none                    | 40         |               |               |
| 13 | female | group B        | high school                 | standard     | none                    | 65         |               |               |
| 14 | male   | group A        | some college                | standard     | completed               | 78         |               |               |
| 15 | female | group A        | master's degree             | standard     | none                    | 50         |               |               |
| 16 | female | group C        | some high school            | standard     | none                    | 69         |               |               |
| 17 | male   | group C        | high school                 | standard     | none                    | 88         |               |               |
| 18 | female | group B        | some high school            | free/reduced | none                    | 18         |               |               |
| 19 | male   | group C        | master's degree             | free/reduced | completed               | 46         |               |               |
| 20 | female | group C        | associate's degree          | free/reduced | none                    | 54         |               |               |

## 3. Transforming Categorical Attributes

### 3.1 Applying Capitalize Each Word

1. Right-click the header of the **gender** column.
2. Choose **Transform column** → **Text transforms** → **Capitalize Each Word**.
3. Repeat the same command for the **race/ethnicity** column.

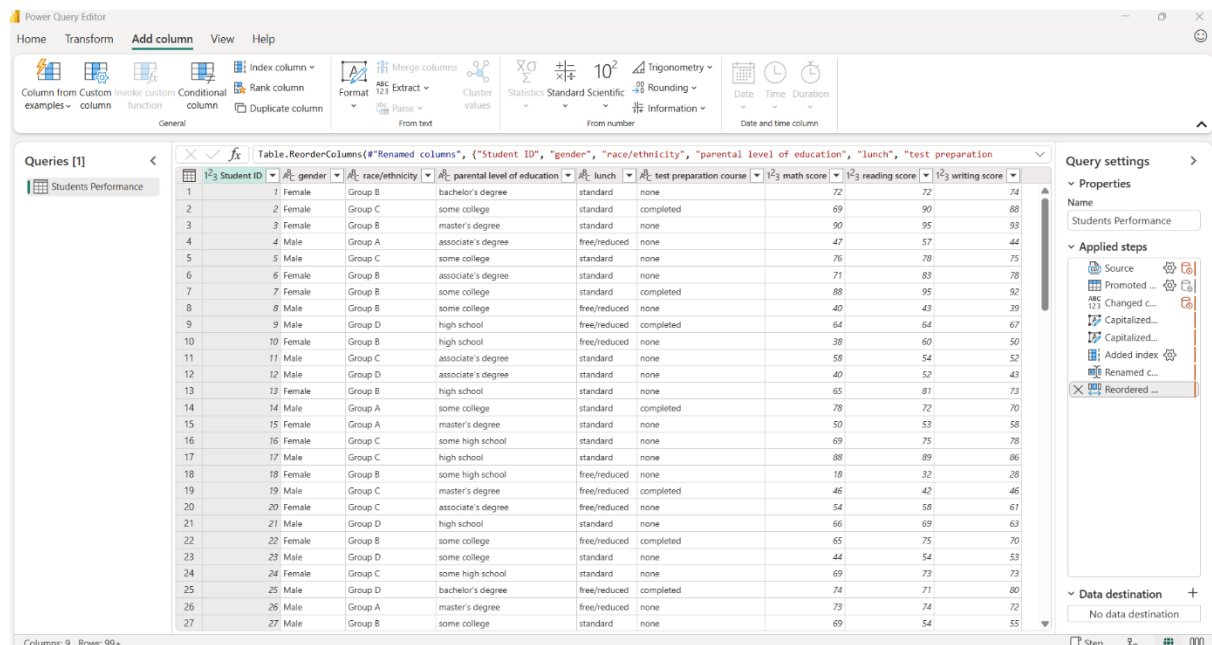


This shows the context menu of the gender column. The **Text transforms** sub-menu offers lowercase, UPPERCASE, Capitalize Each Word, **Trim** (removes leading and trailing spaces), **Clean** (removes non-printing characters), Length, JSON and XML. Capitalize Each Word was selected so that “female” becomes “Female” and “group B” becomes “Group B”. Trim and Clean are good additional safeguards for any text column that has been typed manually or exported from another system.

## 4. Cleaning the Dataset

### 4.1 Creating a Unique Student ID

The original file does not contain an identifier. Because the later steps (unpivoting and building relationships) need a reliable key, an index column was added with Add Column → Index Column → From 1. The new column was renamed Student ID and moved to the first position. Each student therefore has a unique, stable number that links the main table to the unpivoted marks table and the details table. The Student ID column is visible as the first column.



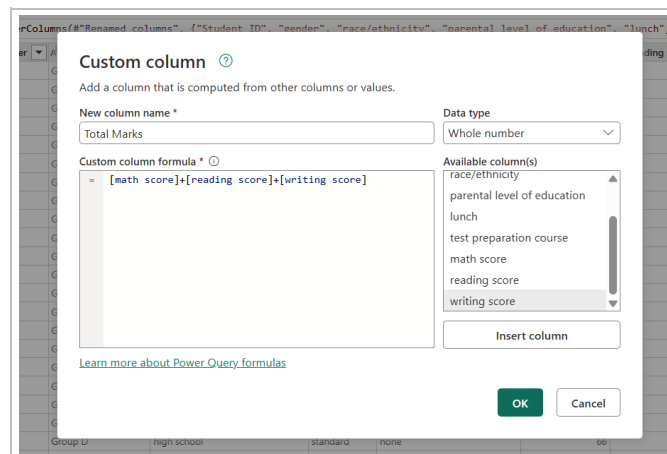
## 5. Creating Calculated Columns in Power Query

Three calculated columns were added to describe each student's overall performance: **Total Marks**, **Average Mark** and **Grade**. Creating them in Power Query (rather than as DAX calculated columns) means the values are computed once when the data is loaded and are available for every later transformation.

### 5.1 Total Marks

The Total Marks column adds the three subject scores. It was created with Add Column → Custom Column, using the name Total Marks, the data type Whole number and the formula shown in this Figure .

```
= [math score] + [reading score] + [writing score]
```

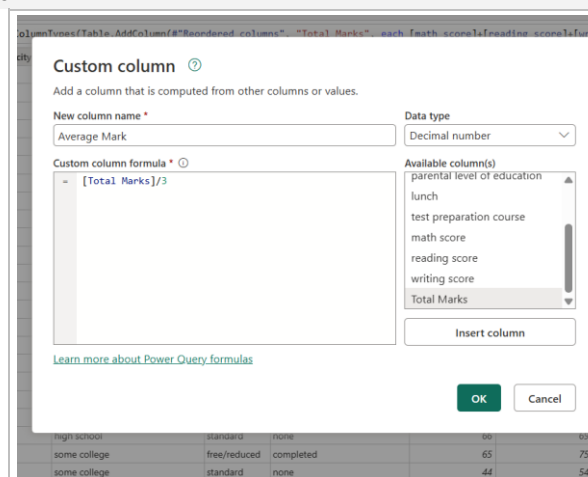


The formula bar at the top of the screenshot shows that the step was applied to the previous step, **Renamed columns**. Because the calculation is performed row by row, each student receives the sum of their own three scores.

### 5.2 Average Mark

The Average Mark is the Total Marks divided by the number of subjects (three). The column was created with a second Custom Column using the **Whole number** data type so that no precision is lost when the division does not give a whole number.

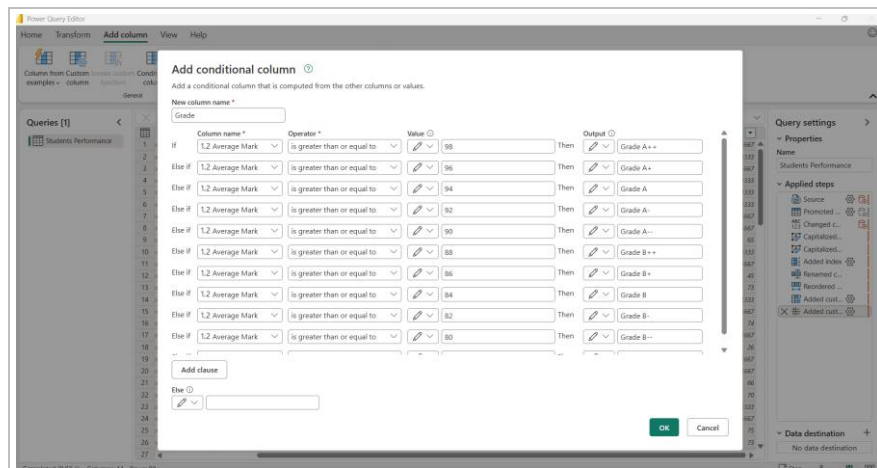
```
= [Total Marks] / 3
```



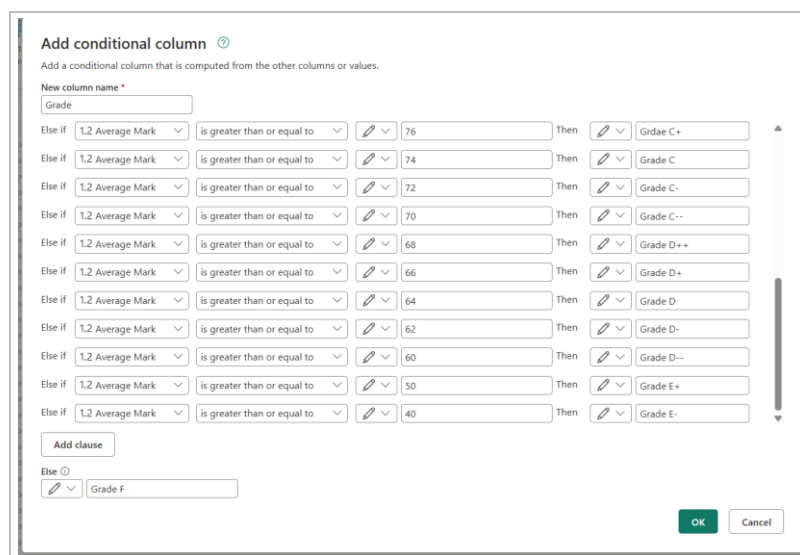
In the final table the Average Mark appears as a whole number . This is because a later Changed type step converts the column to Int64, which rounds the value to the nearest integer for display and for the Pass/Fail test. The Grade, however, was calculated *before* this rounding, on the exact decimal average

### 5.3 Performance Grade (Conditional Logic)

The Grade column was created with **Add Column** → **Conditional Column**. The dialog evaluates a list of **If / Else if** clauses from the top down and returns the output of the first clause that is true; the **Else** value is returned when none of the conditions is met. Every clause tests whether **Average Mark** is *greater than or equal to* a threshold.



The first clauses define the high-performance bands. An average of 98 or more receives Grade A++, 96 or more Grade A+, 94 or more Grade A, and so on in steps of two marks down to Grade B-- for an average of 80 or more.



This shows the lower part of the rule list. The bands continue in steps of two marks through the C and D families (76 down to 60). Below 60 the scale becomes wider: **Grade E+** for 50 or more and **Grade E-** for 40 or more. The **Else** clause assigns **Grade F** to every student whose average is below 40.

| Grade | Min. average | Grade | Min. average | Grade  | Min. average  |
|-------|--------------|-------|--------------|--------|---------------|
| A++   | 98           | B     | 84           | D++    | 68            |
| A+    | 96           | B-    | 82           | D+     | 66            |
| A     | 94           | B--   | 80           | D      | 64            |
| A-    | 92           | C+    | 76           | D-     | 62            |
| A--   | 90           | C     | 74           | D--    | 60            |
| B++   | 88           | C-    | 72           | E+     | 50            |
| B+    | 86           | C--   | 70           | E- / F | 40 / below 40 |

The procedure asks for four broad performance levels. The detailed grade scale used here refines those levels as shown in this Table.

| Grade family        | Average Mark range | Performance level         |
|---------------------|--------------------|---------------------------|
| A (A++ to A--)      | 90 and above       | High performance          |
| B (B++ to B--)      | 80 to below 90     | Good performance          |
| C (C+ to C--)       | 70 to below 80     | Average performance       |
| D and E (D++ to E-) | 40 to below 70     | Below average performance |
| F                   | Below 40           | Low performance           |

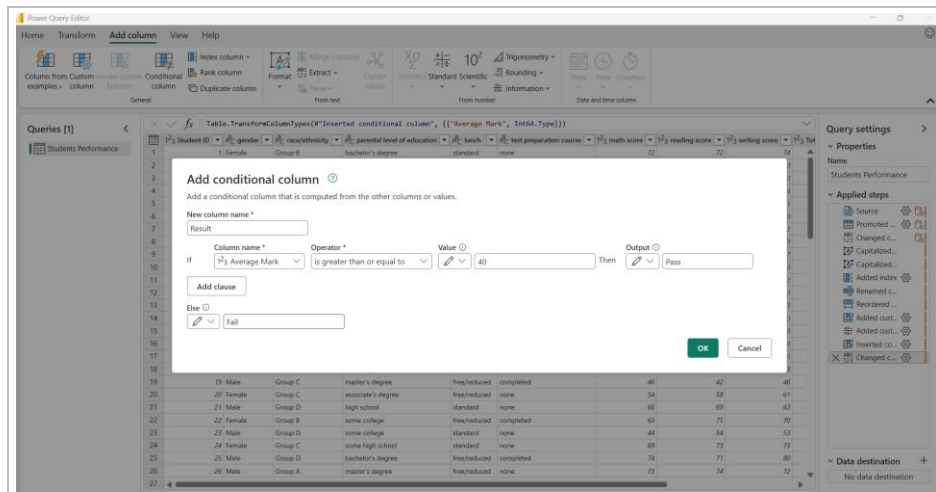
## 6. Creating the Pass/Fail Classification

### 6.1 Pass Condition

A student is considered to have **passed** when the Average Mark is **40 or higher**, and to have **failed** otherwise. The threshold of 40 was chosen because it matches the lowest passing grade band in the grading scale (Grade E-): a student who earns Grade F is below 40 and therefore fails. Keeping the pass mark consistent with the grade scale avoids contradictory classifications.

### 6.2 Implementation

1. Select **Add Column → Conditional Column**.
2. Enter the new column name **Result**.
3. Set the rule **If Average Mark is greater than or equal to 40 Then Pass**.
4. Set the **Else** value to **Fail** and click **OK**.

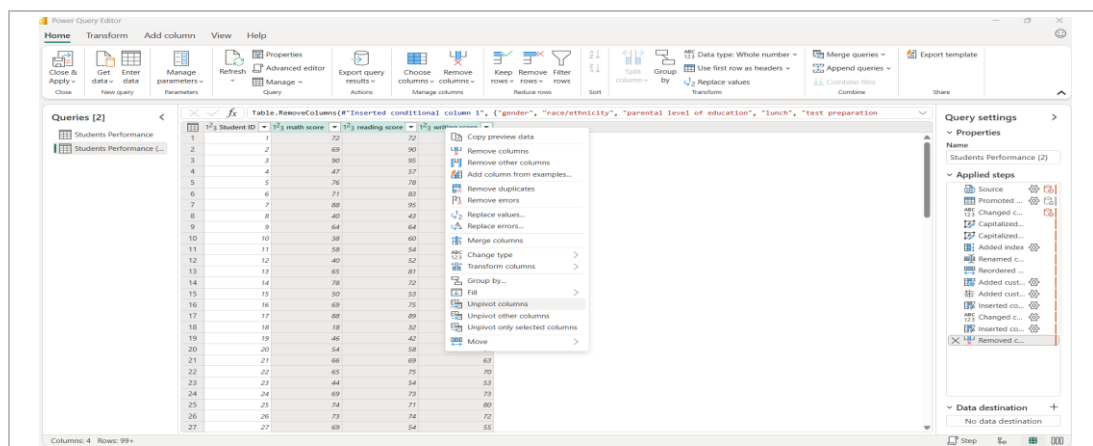


The new column contains only two distinct values, **Pass** and **Fail**, which makes it ideal for slicers and for the Overall Pass % measure created .

## 7. Reshaping Subject Data with Unpivot

### 7.1 Preparing a Separate Query

To keep the main table intact, a second query, **Students Performance (2)**, was created from the first. In this query the **Remove columns** command was used to delete every column except **Student ID**, **math score**, **reading score** and **writing score**.



### 7.2 Applying the Unpivot Transformation

Right-clicking a column header open

- **Unpivot columns** – unpivots the columns currently selected (for example the three score columns).

Student ID was kept fixed and all other columns were unpivoted. The generated step is shown in the formula bar.

```
= Table.UnpivotOtherColumns("#Removed columns", {"Student ID"}, "Attribute", "Value")
```

## 8. Preparing the Data Model

### 8.1 Removing Unnecessary Columns

The **Students Details** table (named *Students Detail* when first created in Power Query and *Students Details* in the loaded model) was derived from the main query by removing the three raw score columns, because those values already live in the unpivoted Students Marks table. The step recorded in the formula bar.

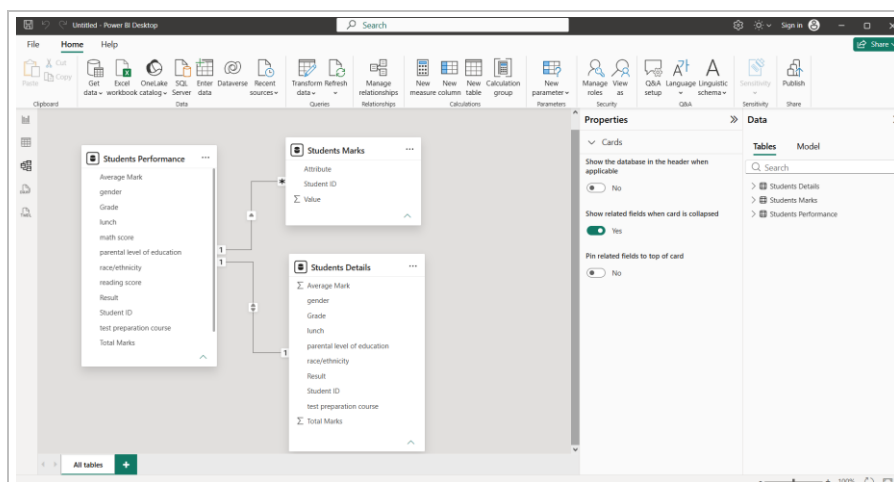
```
= Table.RemoveColumns(#"Inserted conditional column 1", {"math score",  
"reading score", "writing score"})
```

| Student ID | gender | race/ethnicity | parental level of education | lunch        | test preparation course | Total Marks | Average Mark | Grade    | Result |
|------------|--------|----------------|-----------------------------|--------------|-------------------------|-------------|--------------|----------|--------|
| 1          | Female | Group B        | bachelor's degree           | standard     | none                    | 238         | 79           | Grade C- |        |
| 2          | Female | Group C        | some college                | standard     | completed               | 247         | 82           | Grade B- |        |
| 3          | Female | Group B        | master's degree             | standard     | none                    | 238         | 79           | Grade A- |        |
| 4          | Male   | Group A        | associate's degree          | free/reduced | none                    | 148         | 49           | Grade E- |        |
| 5          | Male   | Group C        | some college                | standard     | none                    | 229         | 76           | Grade C+ |        |
| 6          | Female | Group B        | associate's degree          | standard     | none                    | 232         | 77           | Grade C+ |        |
| 7          | Female | Group B        | some college                | standard     | completed               | 275         | 92           | Grade A+ |        |
| 8          | Male   | Group B        | some college                | free/reduced | none                    | 122         | 41           | Grade E- |        |
| 9          | Male   | Group D        | high school                 | free/reduced | completed               | 195         | 65           | Grade D  |        |
| 10         | Female | Group B        | high school                 | free/reduced | none                    | 148         | 49           | Grade E- |        |
| 11         | Male   | Group C        | associate's degree          | standard     | none                    | 164         | 55           | Grade E+ |        |
| 12         | Male   | Group D        | associate's degree          | standard     | none                    | 135         | 45           | Grade E- |        |
| 13         | Female | Group A        | high school                 | standard     | none                    | 279         | 93           | Grade C- |        |
| 14         | Male   | Group A        | some college                | standard     | completed               | 200         | 73           | Grade C- |        |
| 15         | Female | Group A        | master's degree             | standard     | none                    | 167         | 56           | Grade E+ |        |
| 16         | Female | Group C        | some high school            | standard     | none                    | 232         | 76           | Grade C  |        |
| 17         | Male   | Group C        | high school                 | standard     | none                    | 263         | 88           | Grade B+ |        |
| 18         | Female | Group B        | some high school            | free/reduced | none                    | 78          | 26           | Grade F  |        |
| 19         | Female | Group C        | master's degree             | free/reduced | completed               | 134         | 45           | Grade E- |        |
| 20         | Female | Group C        | associate's degree          | free/reduced | none                    | 173         | 58           | Grade E+ |        |
| 21         | Male   | Group D        | high school                 | standard     | none                    | 198         | 66           | Grade D+ |        |
| 22         | Female | Group B        | some college                | free/reduced | completed               | 270         | 90           | Grade C- |        |
| 23         | Male   | Group D        | some college                | standard     | none                    | 151         | 50           | Grade E+ |        |
| 24         |        |                |                             |              |                         |             |              |          |        |

The status bar reports **Columns: 10**: Student ID, gender, race/ethnicity, parental level of education, lunch, test preparation course, Total Marks, Average Mark, Grade and Result. Removing other columns .

### 8.4 Keys and Relationships

The **Student ID** column is the key that links the tables. Figure shows the **Model view** after the data was loaded, with the relationships created between the tables.



## 9. Loading the Data

When all transformations were complete, the data was loaded into the Power BI model with the following steps:

1. In the Power Query Editor, select **Home** → **Close & Apply**.
2. Power BI applies every query, loads the tables into the model and returns to the main window.
3. Open the **Data view** and **Model view** to review the loaded tables.
4. Check that the calculated columns (Total Marks, Average Mark, Grade, Result) and the unpivoted data appear correctly.

After loading, the **Data pane** lists the three tables **Students Details**, **Students Marks** and **Students Performance**. The model view displays the fields of each table, with the calculated columns Total Marks and Average Mark shown as numeric fields ( $\Sigma$ ) and Grade and Result as text fields. The loaded model is therefore ready for the measures defined in the next section.

| Student ID | gender | race/ethnicity | parental level of education | lunch        | test preparation course | math score | reading score | writing score |
|------------|--------|----------------|-----------------------------|--------------|-------------------------|------------|---------------|---------------|
| 1          | Female | Group B        | bachelor's degree           | standard     | none                    | 72         | 72            | 74            |
| 2          | Female | Group C        | some college                | standard     | completed               | 69         | 90            | 88            |
| 3          | Female | Group B        | master's degree             | standard     | none                    | 90         | 95            | 93            |
| 4          | Male   | Group A        | associate's degree          | free/reduced | none                    | 47         | 57            | 44            |
| 5          | Male   | Group C        | some college                | standard     | none                    | 76         | 78            | 75            |
| 6          | Female | Group B        | associate's degree          | standard     | none                    | 77         | 83            | 78            |
| 7          | Female | Group B        | some college                | standard     | completed               | 88         | 95            | 92            |
| 8          | Male   | Group B        | some college                | free/reduced | none                    | 40         | 43            | 39            |
| 9          | Male   | Group D        | high school                 | free/reduced | completed               | 64         | 64            | 67            |
| 10         | Female | Group B        | high school                 | free/reduced | none                    | 38         | 60            | 50            |
| 11         | Male   | Group C        | associate's degree          | standard     | none                    | 58         | 54            | 52            |
| 12         | Male   | Group D        | associate's degree          | standard     | none                    | 40         | 52            | 43            |
| 13         | Female | Group B        | high school                 | standard     | none                    | 65         | 81            | 73            |
| 14         | Male   | Group A        | some college                | standard     | completed               | 78         | 72            | 70            |
| 15         | Female | Group A        | master's degree             | standard     | none                    | 50         | 53            | 58            |
| 16         | Female | Group C        | some high school            | standard     | none                    | 69         | 75            | 78            |
| 17         | Male   | Group C        | high school                 | standard     | none                    | 88         | 89            | 86            |
| 18         | Female | Group B        | some high school            | free/reduced | none                    | 18         | 32            | 28            |
| 19         | Male   | Group C        | master's degree             | free/reduced | completed               | 46         | 42            | 46            |
| 20         | Female | Group C        | associate's degree          | free/reduced | none                    | 54         | 58            | 61            |
| 21         | Male   | Group D        | high school                 | standard     | none                    | 66         | 69            | 62            |
| 22         | Female | Group B        | some college                | free/reduced | completed               | 65         | 75            | 70            |
| 23         | Male   | Group D        | some college                | standard     | none                    | 44         | 54            | 53            |
| 24         | Female | Group C        | some high school            | standard     | none                    | 69         | 73            | 73            |
| 25         | Male   | Group D        | bachelor's degree           | free/reduced | completed               | 74         | 71            | 80            |
| 26         | Male   | Group A        | master's degree             | free/reduced | none                    | 73         | 74            | 72            |

## 10. Creating DAX Measures

### 10.1 Creating a Dedicated Measures Table

To keep the field list organised, all measures were stored in a separate table named **Measures Table**. It was created using **Home** → **Enter data**, which opens the *Get Data (Power Query)* window with a **Blank table**. The table was named **Measures Table** and loaded (Figure 12). The empty default column (Column1) can be hidden or deleted once the measures exist.

Get Data (Power Query)

Connect to data source

Blank table

Copy and paste data into the table, or enter data manually.

Use first row as headers

| Column1 |
|---------|
|         |

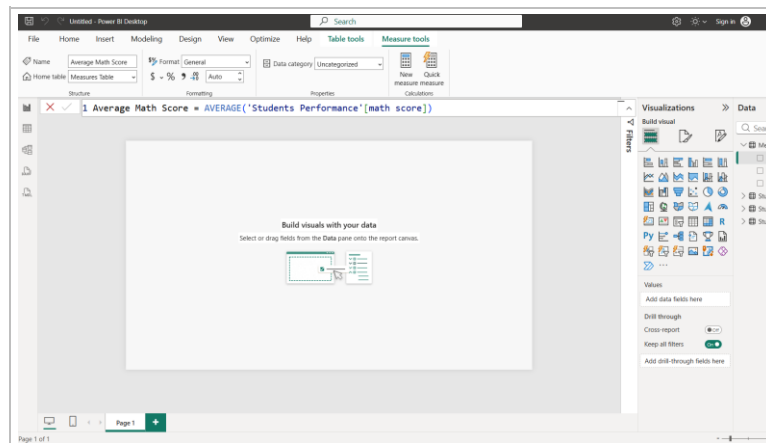
Name: Measures Table

Back Cancel Next

## 10.2 Average Math Score

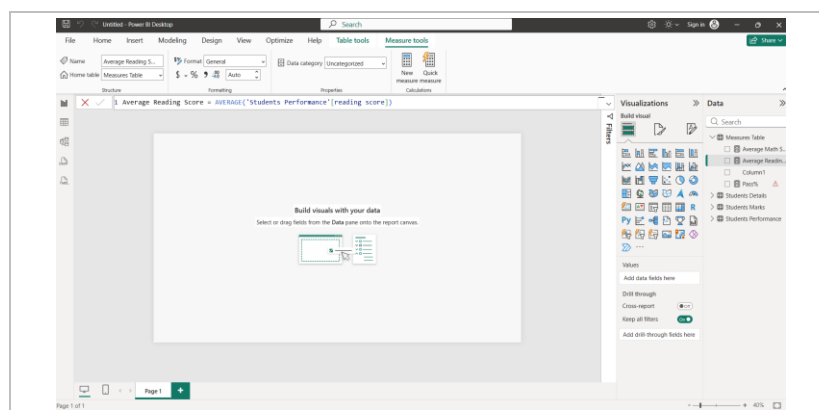
The first measure calculates the arithmetic mean of the math score column. It is created via **Home** → **New measure** (or Measure tools → New measure), with the **Home table** set to Measures Table.

```
Average Math Score = AVERAGE('Students Performance'[math score])
```



## 10.3 Average Reading Score

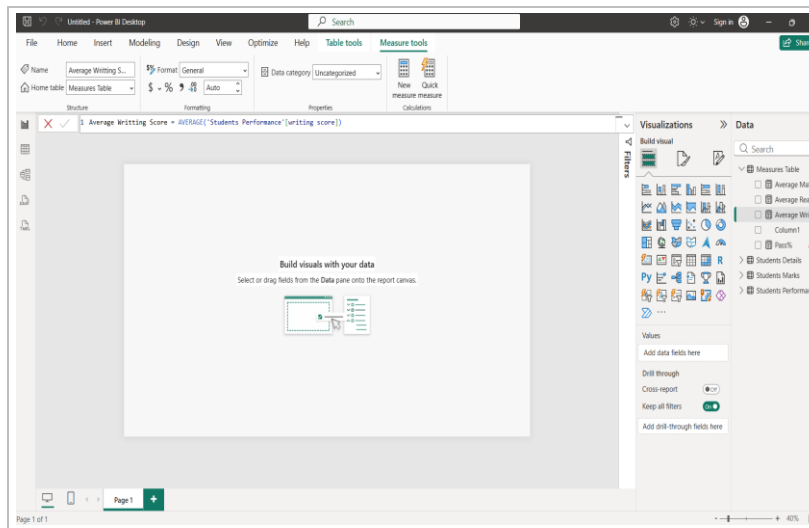
```
Average Reading Score = AVERAGE('Students Performance'[reading score])
```



## 10.4 Average Writing Score

```
Average Writting Score = AVERAGE('Students Performance'[writing score])
```

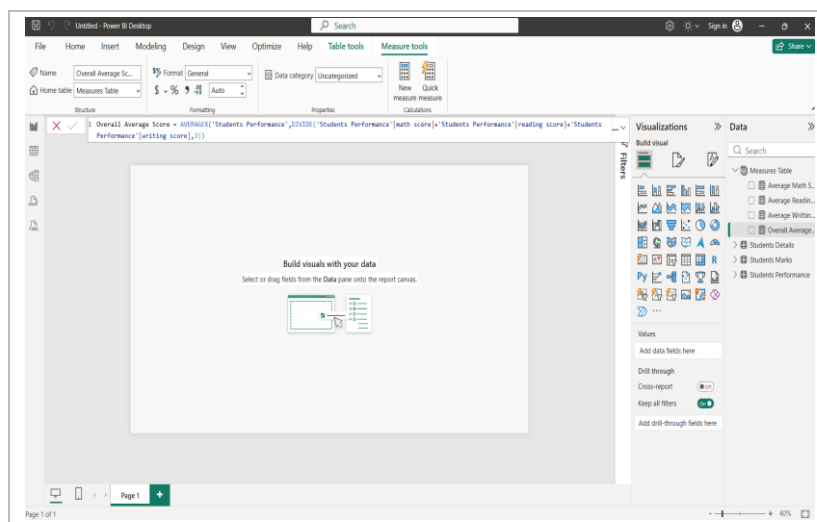
a



## 10.5 Overall Average Score

The Overall Average Score gives the mean of each student's three-subject average. **AVERAGEX** iterates over the Students Performance table, **DIVIDE** calculates the average of the three scores for each student (and safely handles division problems), and AVERAGEX then averages those per-student results.

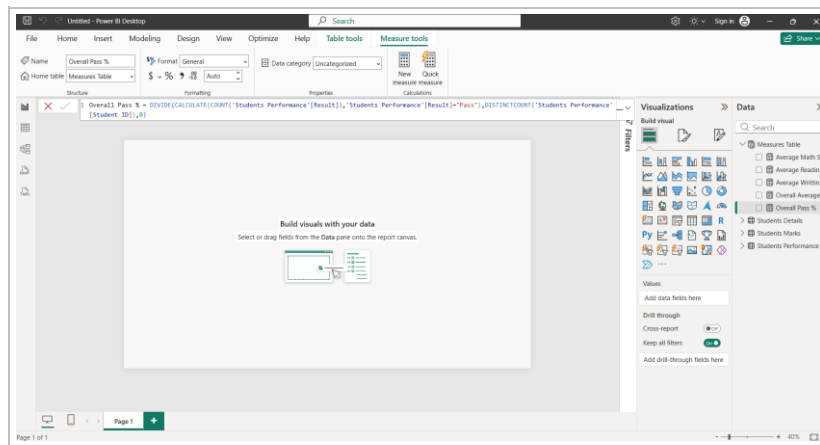
```
Overall Average Score =
AVERAGEX (
    'Students Performance',
    DIVIDE('Students Performance'[math score]
        + 'Students Performance'[reading score]
        + 'Students Performance'[writing score], 3)
)
```



## 10.6 Overall Pass %

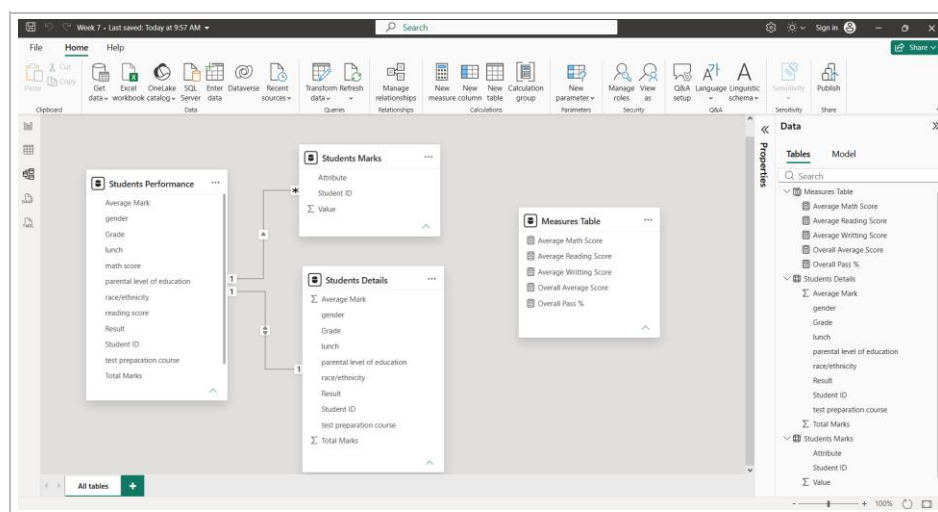
The Overall Pass % measure counts the students whose Result is "Pass" and divides by the number of distinct students:

```
Overall Pass % =
DIVIDE (
    CALCULATE(COUNT('Students Performance'[Result]),
        'Students Performance'[Result] = "Pass"),
    DISTINCTCOUNT('Students Performance'[Student ID]),
    0
)
```



## 11. Validating the Results

The final stage compares the transformed data with the original dataset and verifies the calculated columns. The completed data model, including the Measures Table with all five measures, is shown in Figure .



### 11.1 Checking the Calculated Scores

Seven students were selected and their results were recomputed by hand from the original scores. Table shows that every Total Marks, Average Mark, Grade and Result agrees with the values produced by Power Query . The exact average is shown as well as the rounded average, because the Grade is assigned from the exact value before rounding.

| Student ID | Math | Reading | Writing | Total Marks | Exact avg. | Average Mark | Grade    | Result |
|------------|------|---------|---------|-------------|------------|--------------|----------|--------|
| 1          | 72   | 72      | 74      | 218         | 72.67      | 73           | Grade C- | Pass   |
| 2          | 69   | 90      | 88      | 247         | 82.33      | 82           | Grade B- | Pass   |
| 3          | 90   | 95      | 93      | 278         | 92.67      | 93           | Grade A- | Pass   |
| 4          | 47   | 57      | 44      | 148         | 49.33      | 49           | Grade E- | Pass   |

| Student ID | Math | Reading | Writing | Total Marks | Exact avg. | Average Mark | Grade     | Result |
|------------|------|---------|---------|-------------|------------|--------------|-----------|--------|
| 7          | 88   | 95      | 92      | 275         | 91.67      | 92           | Grade A-- | Pass   |
| 8          | 40   | 43      | 39      | 122         | 40.67      | 41           | Grade E-  | Pass   |
| 18         | 18   | 32      | 28      | 78          | 26.00      | 26           | Grade F   | Fail   |

## 11.2 Verification Checklist

| Verification item                        | Method                                          | Status   |
|------------------------------------------|-------------------------------------------------|----------|
| Data types are correct                   | Header icons; Changed type step                 | Verified |
| No missing values or errors              | Column quality (100% valid, 0% error, 0% empty) | Verified |
| Categorical values standardised          | Female/Male and Group A–D in the data grid      | Verified |
| Total Marks and Average Mark are correct | Manual recalculation (Table 12)                 | Verified |
| Grade assigned from correct band         | Compared with the thresholds in Table 7         | Verified |
| Pass/Fail follows the 40-mark rule       | Student 18 fails; all others pass               | Verified |
| Unpivoted values match original scores   | Table 9 compared with Figures 1 and 2           | Verified |
| Relationships and measures are in place  | Model view and Data pane (Figures 11 and 18)    | Verified |

## 13. Conclusion

In this task the **Student Performance in Exams** dataset was imported into Power BI Desktop and prepared entirely in the Power Query Editor. The data was inspected for missing values, errors and duplicates, the categorical attributes gender and race/ethnicity were standardised, and a unique **Student ID** key was created. New attributes were derived: **Total Marks**, **Average Mark**, a multi-level **Grade** and a **Pass/Fail Result** (pass mark 40). The subject scores were reshaped with **Unpivot**, and the queries were organised into the Students Performance, Students Marks and Students Details tables linked by Student ID.

After *Close & Apply*, five DAX measures were created in a dedicated Measures Table and the calculated values were validated against the original scores. The refinements listed in Section 12 can be applied to finalise the model.